

The camera already saw it.

**A single frame cannot
tell you whether a box
was placed or dropped.
That lives in a
sequence.**

Warehouses record everything and watch none of it. CCTV produces evidence *after* a damage claim, by which point the product is broken, the cause is disputed, and the handling pattern that caused it has repeated unnoticed for weeks. DockSense turns loading-bay video into reviewable incidents: what happened, which entity, when, why it was judged risky, and how sure the system is.

4.2 implemented, fixed by evidence

More on our website

Apple M2 / 8 GB network: off442 needed

Track the entity, then reason over its short history.

video	perception	tracking	temporal features	12 behaviours	event graph	risk	incident	console
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The highlighted stages are the ones that decide whether a movement was a drop, a throw, a drag or a careful placement. They cost **under one millisecond per frame** — detection is ~99% of the pipeline, so the part that differentiates this system is effectively free.

THRESHOLDS IN OBJECT-HEIGHTS

A box falling 1.5× its own height fell the same amount whether the camera is 3 m or 8 m away. **Pixel thresholds silently stop firing the moment a camera moves** — and nobody notices, because the failure is silence.

RISK AND CONFIDENCE NEVER MERGE

Risk is **how bad if this is real**. Confidence is **how sure we are it is real**. Multiplying them into one number hides the case that matters most: high risk, low confidence — which belongs in a review queue, not an alarm.

EVERY DETECTOR HAS A NAMED HARD NEGATIVE

Gentle placement must not fire drop. Carrying at knee height must not fire drag. **A detector that fires on everything passes every positive test** and destroys the demo.

Twelve behaviours, one offline command.

DETECTION WITHOUT TRAINING

YOLO-World open-vocabulary: warehouse classes come from text prompts in a config file, so **the prompt strings are a tuning surface** — as much a threshold as anything else. No model was trained; none could be, on 8 GB and no CUDA.

OFFLINE BY CONSTRUCTION

Weights are committed; CLIP ships as four 90 MiB chunks reassembled with a SHA256 check, because it exceeds GitHub's per-file limit. **Verified with socket.connect patched to raise** — zero outbound connections.

BEHAVIOUR, NOT IDENTITY

Stored evidence clips blur the upper quarter of every detected person box. Deliberately **not** face detection — running a face detector to decide what to blur builds the exact capability we say we don't have. It reduces identifiability; it is not anonymisation.

BEHAVIOURS, TIERED BY THE EVIDENCE BEHIND THEM — NOT BY AMBITION		
TIER	BEHAVIOURS	WHAT THE TIER MEANS
Strong	B01 drop · B05 improper stack · B06 unstable stack · B07 zone	Held-out precision and recall 1.000 on synthetic reasoning tests
Moderate	B02 throw · B03 drag · B09 stepping · B12 unsafe surface	Fires, with named failure modes we can describe
Lightly validated	B04 rough handling · B10 heavy handling · B11 unsafe sequence	Proxy heuristics with confounds we can name, so we name them

B10 is called *“large item handled without equipment present”*, not “unsafe lift”. It cannot see mass, and mass is the whole concept, so the name says what was observed rather than what was inferred.

Remove tracking and the system goes completely silent.

Measured on physics-rendered clips with the renderer's exact box geometry injected as perception, so this isolates the reasoning layer. Two splits: a tune set of 9 clips development happened against, and a heldout set of 27 drawn separately — different release heights, launch speeds, stack offsets, and carton sizes from 60 to 100 px.

REASONING-LAYER F1 — MECHANISMS SWITCHED OFF ONE AT A TIME			
VARIANT	TUNE (N=9)	HELDOUT (N=27)	READING
baseline	1.000	0.933	The gap between columns is the size of the overfit
no_tracking	0.000	0.000	Nothing fires at all. Every behaviour is defined over a sequence.
no_smoothing	1.000	0.867	Costs nothing on fixed clips; costs 0.066 once sizes and speeds vary
no_event_graph	0.933	0.933	No delta — it feeds the risk score, not event detection. Reported anyway.

WHY THE TUNE COLUMN READS 1.000

Because thresholds were changed in response to failures on those nine clips. **A perfect score on your own tuning set is a warning, not a result** — so the held-out split exists, and 0.933 is the number we quote.

WHAT A HELD-OUT SPLIT DOESN'T FIX

Both splits come from the same renderer. A systematic error in how it models handling appears identically in both. **Only real footage closes that gap** — which is the next slide.

Then we pointed it at a real dock, and it did badly.

1/9

expected behaviours reported across seven clips of genuine loading-bay operations. Not silent — incidents on 5 of 7 clips — but mostly the two loosest detectors rather than the behaviour each clip was recorded to show.

This is the number we lead with, because everything on the previous slide was measured on clips that could not fail in these particular ways. One afternoon of real footage found four defects:

FIXED

A config knob was declared and read by nothing.

`min_track_age_frames` sat in the config file, documented, wired to nothing. A one-frame-old track's first velocity is a full-magnitude jump — on soft footage that jump is detector jitter, and it was firing *throw* on clips of people dragging.

FIXED

A single horizontal floor line cannot describe a perspective view.

One line puts a box on the floor at one depth and a metre in the air at another. That is why drag never fired on any of four dragging clips. The nearest person's feet are now the local ground reference — calibration-free, and correct at every depth.

Four things move this further than any amount of tuning.

01 Sixty seconds of ordinary handling

One clip where nothing bad happens. Without it, precision on real footage stays permanently unmeasurable and the alarm rates cannot be interpreted at all. It is the cheapest item on this list and the most valuable.

02 Start and end times logged at record time

Turns a clip-level presence check into real precision and recall. Reconstructing spans afterwards means scoring the system against labels drawn after seeing what the system did.

03 Direct camera export

Session 1 is a phone recording of a CCTV playback window. Exporting from the recorder removes two lossy encodes, the screen moiré, the software chrome and the annotations drawn on the footage — in one step.

04 Closer framing

A carton is currently ~200 px tall. Raise that and the acceleration signal rises above box jitter, which is the one thing standing between throw and drag being separable at all.

Everything claimed here is traceable to a file: the claims ledger in `STATE.md` lists each number, where it was measured, and which ones are still marked *not measured*.